

Problem set 3

1. (Filming a movie). A producer (Polly), an operator (Oliver) and a composer (Chris) are filming a movie. The movie yields \$10000 profit. Oliver himself can earn \$2000 by filming a wedding, Chris himself can earn \$3000 by selling the music, Polly herself can earn only \$1000 by event management. Oliver and Chris can earn \$6000 by creating a music film for a wedding. Oliver and Polly can earn \$4000 by organizing the whole wedding. Polly and Chris can earn \$4500 by organizing a party.

- (a) Formalize the situation as a coalitional game. Is it superadditive? Supermodular?
- (b) Find the core and draw it on the plane (Polly payoff, Chris payoff).
- (c) Find the Shapley value. Does it lie inside the core?

Solution.

Let

$$N = \{P, O, C\},$$

where P is Polly, O is Oliver, and C is Chris.

The payoff function is

$$\begin{aligned} v(\emptyset) &= 0, \\ v(P) &= 1000, & v(O) &= 2000, & v(C) &= 3000, \\ v(PO) &= 4000, & v(PC) &= 4500, & v(OC) &= 6000, \\ v(POC) &= 10000 \end{aligned}$$

The game is superadditive, since every coalition gives at least as much as separate coalitions. Showing this:

$$\begin{aligned} v(PO) &= 4000 \geq v(P) + v(O) = 1000 + 2000 = 3000, \\ v(PC) &= 4500 \geq v(P) + v(C) = 1000 + 3000 = 4000, \\ v(OC) &= 6000 \geq v(O) + v(C) = 2000 + 3000 = 5000, \\ v(POC) &= 10000 \geq v(P) + v(OC) = 1000 + 6000 = 7000, \\ v(POC) &= 10000 \geq v(O) + v(PC) = 2000 + 4500 = 6500, \\ v(POC) &= 10000 \geq v(C) + v(PO) = 3000 + 4000 = 7000 \end{aligned}$$

Hence the game is superadditive.

Show supermodularity:

$$v(POC) + v(P) \geq v(PO) + v(PC),$$

because

$$10000 + 1000 = 11000 \geq 4000 + 4500 = 8500$$

Similarly,

$$v(POC) + v(O) \geq v(PO) + v(OC),$$

because

$$10000 + 2000 = 12000 \geq 4000 + 6000 = 10000$$

And

$$v(POC) + v(C) \geq v(PC) + v(OC),$$

because

$$10000 + 3000 = 13000 \geq 4500 + 6000 = 10500$$

So, the game is supermodular.

The core consists of allocations

$$x = (x_P, x_O, x_C)$$

such that

$$x_P + x_O + x_C = 10000,$$

and

$$x_P \geq 1000, \quad x_O \geq 2000, \quad x_C \geq 3000,$$

$$x_P + x_O \geq 4000,$$

$$x_P + x_C \geq 4500,$$

$$x_O + x_C \geq 6000$$

To draw the core we use

$$x_O = 10000 - x_P - x_C$$

Then the constraints become

$$1000 \leq x_P \leq 4000,$$

$$3000 \leq x_C \leq 6000,$$

$$4500 \leq x_P + x_C \leq 8000$$

Hence the core is a hexagon with vertices

$$(1000, 3500),$$

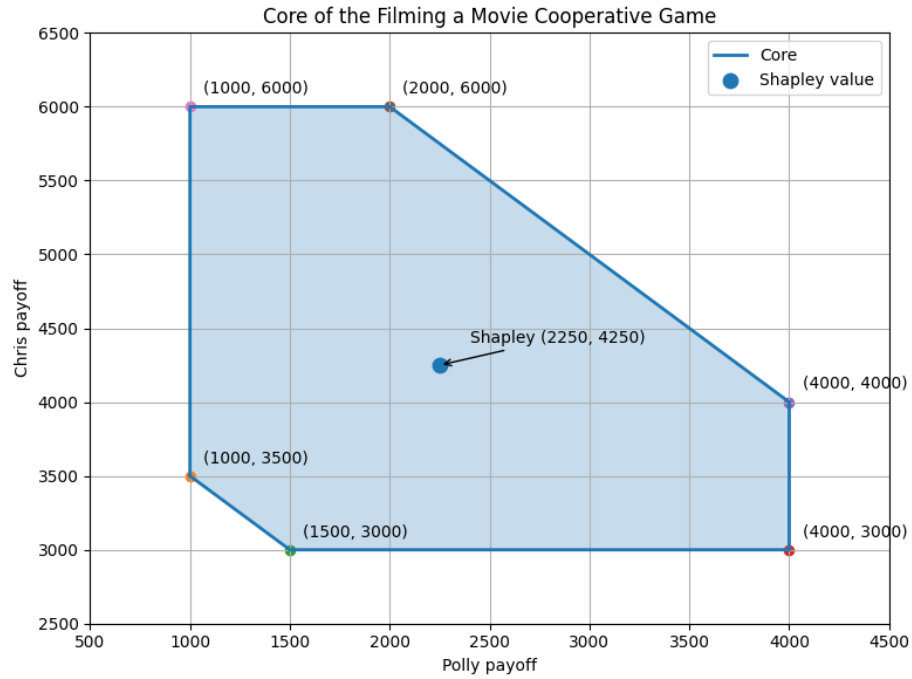
$$(1500, 3000),$$

$$(4000, 3000),$$

$$(4000, 4000),$$

$$(2000, 6000),$$

$$(1000, 6000)$$



Now compute the Shapley value:

$$sv_P = \frac{1000 + 1000 + 2000 + 4000 + 1500 + 4000}{6} = 2250,$$

$$sv_O = \frac{3000 + 5500 + 2000 + 2000 + 5500 + 3000}{6} = 3500,$$

$$sv_C = \frac{6000 + 3500 + 6000 + 4000 + 3000 + 3000}{6} = 4250$$

Order	P	O	C
POC	1000	3000	6000
PCO	1000	5500	3500
OPC	2000	2000	6000
OCP	4000	2000	4000
CPO	1500	5500	3000
COP	4000	3000	3000
Shapley value	2250	3500	4250

Hence

$$sv(P, O, C) = (2250, 3500, 4250).$$

Check whether it lies in the core:

$$2250 + 3500 + 4250 = 10000,$$

$$2250 \geq 1000, \quad 3500 \geq 2000, \quad 4250 \geq 3000,$$

$$2250 + 3500 = 5750 \geq 4000,$$

$$2250 + 4250 = 6500 \geq 4500,$$

$$3500 + 4250 = 7750 \geq 6000$$

Thus all core constraints are satisfied, so

$$\text{sv}(P, O, C) = (2250, 3500, 4250) \in \text{Core}(v)$$

Therefore,

the game is superadditive and supermodular,

and

the Shapley value lies inside the core